Aim

To study, implement, and analyze commonly used TCP/IP utility commands.

Theory

TCP/IP (Transmission Control Protocol/Internet Protocol) is the standard communication protocol used for networking and the Internet. Various TCP/IP utility commands help users diagnose network connectivity, verify configurations, resolve domain names, and trace communication paths.

Common TCP/IP Utility Commands

1. **ping** – Checks connectivity between two hosts.
2. **ipconfig** (Windows) / **ifconfig** or **ip addr** (Linux) – Displays IP configuration.
3. **tracert** (Windows) / **traceroute** (Linux) – Displays the route packets take to a destination.
4. **nslookup** – Finds the IP address associated with a domain name.
5. **netstat** – Displays active network connections and listening ports.
6. **arp** – Displays and modifies the ARP cache.
7. **hostname** – Displays the system's host name.
8. **route** – Displays and modifies the routing table.

Procedure

1. Open Command Prompt

- Press **Win + R**.
- Type **cmd** and press **Enter**.

Command 1: ping

To check whether a remote host is reachable.

Syntax

ping <hostname/IP address>

Example

ping google.com

Observation

- Displays the number of packets sent, received, and lost.
- Shows response time (latency).

Command 2: ipconfig

To display the current network configuration.

Syntax

ipconfig

For detailed information

ipconfig /all

Observation

Displays:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server
- MAC Address (using /all)

Command 3: tracert

To trace the route packets travel to reach a destination.

Syntax

tracert <hostname/IP address>

Example

tracert google.com

Observation

Displays every router (hop) between the source and destination along with the response time.

Command 4: nslookup

To obtain the IP address of a domain name.

Syntax

nslookup <domain_name>

Example

nslookup google.com

Observation

Displays:

- DNS Server
- Domain Name
- IP Address

Command 5: netstat

To display active network connections.

Syntax

netstat

Common Options

netstat -a

Shows all active and listening connections.

netstat -an

Shows numerical addresses and ports.

netstat -r

Displays the routing table.

Observation

Displays:

- Protocol
- Local Address
- Foreign Address
- Connection State

Command 6: arp

To display the Address Resolution Protocol (ARP) cache.

Syntax

arp -a

Observation

Shows:

- IP Address
- Physical (MAC) Address
- Type (Dynamic/Static)

Command 7: hostname

To display the computer's host name.

Syntax

hostname

Observation

Displays the system name registered on the network.

Command 8: route

Aim

To display the routing table.

Syntax

route print